

MATHEW R. THOMSON

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SUMMARY

Data Engineer with 3+ years of experience building and optimizing distributed ETL pipelines on AWS EMR (EC2/EKS) and Snowflake for large-scale financial datasets. Skilled in Spark performance tuning, batch orchestration (Control-M, Jenkins), and historical data warehousing (SCD Type-2). Experienced in integrating ML workflows into production pipelines, enabling scalable ML-powered Data Products.

SKILLS

Core Data Engineering: Spark (PySpark), Snowflake, EMR (EC2/EKS), dbt, Airflow, SCD Type-2, CDC

Cloud & Infrastructure: AWS (S3, EMR, EKS), Docker, Linux

Orchestration & DevOps: Control-M, Jenkins, Git

Languages: Python (Advanced), SQL (Advanced)

Applied ML (Supporting Skillset): TensorFlow, PyTorch, Scikit-Learn

EXPERIENCE

Data Engineer | Capco [Client: Freddie Mac] (Tysons, VA, USA)

Apr 2024 – Present

- Designed and productionized distributed ETL pipelines on AWS EMR (EC2 & EKS), ingesting 10+ diverse datasets and processing 40M+ records per batch using PySpark.
- Optimized Spark workloads, reducing end-to-end batch runtimes from ~40 min to sub 10 min, accelerating daily reporting for senior stakeholders.
- Built dependency-aware daily and monthly batch workflows via Control-M and Jenkins, enforcing SLAs and automated failure recovery.
- Engineered Snowflake data models and curated high-visibility data products for senior VPs and enterprise consumers using advanced SQL (40+ joins, CTEs, window functions).
- Spearheaded Snowpark-based transformation standardization, and Snowflake query optimization, reducing code divergence by 40% and saving \$16K annually.
- Reconstructed 22+ years of historical data with Informatica IICS and Snowflake, establishing consistent baselines and improving reconciliation accuracy.
- Integrated ML-driven feature engineering into batch pipelines, enabling enterprise-scale predictive analytics.
- Collaborated with analytics and business stakeholders to translate complex reporting requirements into scalable, production-ready pipelines influencing key data decisions.

Lead AI Developer | Bosmos (Atlanta, GA, USA)

Sep 2023 – Mar 2024

- Led design and deployment of a production AI chatbot platform using TensorFlow-based NLP pipelines for real-time conversational inference.
- Managed a 5-member engineering team, delivering end-to-end ML model training, deployment, and API integration under Agile practices.
- Optimized backend architectures for ML inference, reducing latency and improving throughput for real-time prediction workloads.

SELECTED RECENT PROJECTS

Batch Analytics Platform | Apache Airflow, dbt, PostgreSQL, Python

Nov 2025 - Feb 2026

- Architected a production-style batch ELT framework with config-driven orchestration simulating enterprise-scale workloads (50M+ record synthetic datasets).
- Designed reusable DAG abstractions enabling plug-and-play dataset onboarding across heterogeneous schemas.
- Validated Spark optimization strategies with dynamic partitioning and shuffle tuning under simulated enterprise-scale workloads.

CDC Historical Warehouse Platform | Python, PostgreSQL, CDC, SCD Type-2

Nov 2025 - Feb 2026

- Built an SCD Type-2 historical warehouse supporting scalable CDC patterns with idempotent batch execution and deterministic replay.
- Implemented versioned record tracking and late-arriving data handling for temporal consistency and auditability.
- Developed deterministic batch reprocessing logic ensuring data integrity across incremental loads.

Data Observability Platform | Python, Data Quality Monitoring, PostgreSQL

Nov 2025 - Feb 2026

- Designed a configuration-driven data observability framework with statistical baselining, freshness validation, and automated anomaly detection.
- Implemented rolling-window metric profiling to proactively detect schema drift, freshness gaps, and SLA violations in distributed batch workloads.

EDUCATION

Carnegie Mellon University, Pittsburgh, PA

May 2023

Master of Science in Mechanical Engineering – Research (ML/CV)

GPA: 4.0/4.0

Vellore Institute of Technology, Vellore (India)

June 2020

Bachelor of Engineering in Mechanical

GPA: 3.8/4.0